

NeuroGen

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Introduction

1.1 Problem Domain

Present Challenge: MRI scans typically generate high-resolution images, but they often show missing or corrupted regions, which negatively affects the reliability of the diagnosis. This makes pinpointing brain tumors extremely difficult.

Essential Requirement: Precise evaluation of glioma tumors is necessary for efficient diagnosis and therapeutic strategizing. Unfortunately, because of the limits of the imaging data, medical experts frequently struggle to provide accurate MRI interpretations.

Study Motivation: By producing upgraded pictures that correct for missing or corrupted regions, this study seeks to solve the deficiencies in MRI analysis for patients diagnosed with gliomas. Our goal is to enhance the accuracy of MRU scans to enable healthcare providers to make well-informed decisions and optimize patient outcomes in the setting of brain tumour management.

1.2 Problem Statement

The project tackles the difficulty of interpreting brain MRI scans with gliomas, where standard algorithms frequently underperform due to the presence of lesions. Such algorithms are usually trained on healthy brain pictures, which limits their usefulness in clinical situations when analysing tumor-affected regions. The proposed program attempts to address this issue by using advanced inpainting techniques to generate healthy 3D brain tissue in tumor-affected areas. This approach will help to restore MRI data integrity, increasing the dependability of essential therapeutic operations including brain tissue segmentation, anatomical parcellation, and tumour growth modelling. The method improves the performance of brain MRI analysis by providing realistic tissue reconstructions, making it better suited for clinical decision making.

1.3 Proposed Solution/Method

The proposed approach is an extensive machine learning method that employs a number of cutting-edge models, including Denoising Diffusion Probabilistic Models (DDPMs), to in-paint (create) 3D healthy brain tissue in the region impacted by the glioma tumour.

We'll use MRI scans of glioma-affected brains as input, together with an in-painting mask to highlight the tumor-affected area. The solution will generate healthy tissue brain within the masked region in voided part of the MRI by experimenting with a variety of in-painting techniques, including DDPMs, Generative Adversarial Networks (GANs), Variational Auto-encoders (VAEs), and possibly Latent Diffusion Models (LDMs), resulting in a complete brain MRI scan that accurately represents a healthy state. This in-painting procedure will aid in the reconstruction of missing or damaged portions with realistic tissue, ultimately boosting the quality of data used in therapeutic jobs.

The training phase will involve T1 MRI modality scans from the BraTs dataset 2024 , which includes voided tumor-affected MRI scans of Glioma-affected patients. The models will learn to synthesize healthy tissue in tumor-affected areas, using the healthy brain tissue mask as input with voided MRI . Model performance will be assessed using conventional metrics such as SSIM, PSNR and MSE to verify that the synthetically generated regions closely mimic real healthy brain tissue. Furthermore, several approaches such as 3D Wavelet-based models or memory efficient 3D models, will be explored to overcome scaling concerns with high-resolution MRIs. The project will not only generate realistic healthy brain tissue, but will also seek to increase the ability to enhance the analysis of brain MRI , making it more useful for clinical applications such as tissue segmentation, anatomical parcellation, and tumour growth tracking. The approach will use both diffusion models and alternative in-painting approaches, resulting in complete results and a strong answer to medical imaging difficulties.

Preliminary Literature Review

2.1 Deep learning-Based 3D inpainting of brain MR images

2.1.1 Summary

This paper [8] addresses the issue of reconstructing 3D brain MR images from sparse 2D slices, which are commonly used due to long scan times for 3D imaging. The authors propose a deep learning model [10] with a U-Net-like structure using DenseNet sub-blocks to perform 3D inpainting. The model was trained using both fidelity loss and perceptual loss, enabling it to recover detailed anatomical structures. They used 680 T1-weighted 3D MR images for training and testing, focusing on assessing the quality of the inpainted images through various metrics like PSNR, SSIM, and HFEN. Results show that the proposed method outperformed conventional interpolation and U-Net models in both image quality and accuracy in morphometric measurements.

2.1.2 Critical Analysis

- 1-Both fidelity and perceptual losses integrated that helped Network to recover more realistic and accurate anatomical structures in comparison to other methods
- 2-Gradient flow and efficient features extraction got better because of usage of DenseNet sub-blocks
- 3-Cortical thickness and gray matter volume which are essential for neuroimaging were effectively recovered
- 4-Boundary artifacts in reconstructed images were a limitation of the model that could potentially impact its clinical applicability. Because the training process uses multiple

layers in the dense block and patch-based learning, the method requires a lot of processing power.

2.1.3 Relation to the project:

Our project and this paper both focuses on reconstructing missing or distorted tissue in 3D MRIs. The use of multi-modal MRI data and the architecture, which combines DenseNet sub-blocks with a U-Net-like model emphasis on 3D in-painting. To target tumor-affected areas, we can modify their methodology by potentially improving the model to incorporate glioma-specific features. Our project aims to expand on the healthy brain tissue recovery discussed in the paper. Specifically, we would concentrate on regions affected by tumors and make sure the algorithm creates realistic healthy tissue

2.2 Conditional Diffusion Models for Semantic 3D Brain MRI Synthesis

2.2.1 Summary:

This paper [3] presents the novel approach to the 3D brain MRI synthesis using conditional diffusion models. Generating the accurate brain MRI synthesis based on labels probabilistic models in denoising and generation of image. They used the dataset of BraTS2021 and generates MRI using segmentation Mask. The main idea is to guide the denoising process through semantic conditioning, to make sure the resulting brain MRI is realistic [13] This paper reviews the already used techniques like GANs and VAEs focusing on its struggle with structural coherence and fine detailing of 3D images, also proposed the ways to overcome these limitations [7] which should be use as a better approach, by integrating sematic guidance which allows the model to produce the high quality MRI scan image anatomically correct and much realistic. The process of training the model uses the 3D brain MRI dataset labelled with brain regions and model iteratively improving the quality. Adapted Vanilla DDM Model for generating 3D volumetric images.

2.2.2 Critical Analysis

1- Used a simple 3D U-NET model.

- 2- Diffusion models are known for their ability to denoise noisy images iteratively improve image quality. Applying this to 3D MRI synthesis ensures that the output is both
- 3- high-resolution and anatomically accurate which is of much importance for clinical applications.
- 4- Earlier models like GANs and VAEs which are not much successful in preserving the anatomical features of images, the models were then conditioned on semantic labels results in more accurate brain structure image generated.
- 5- The reliance on the semantic labels also means that the model's performance can be limited by quality of the dataset and granularity of the label data.
- 6- The diffusion and denoising of the image require multiple iterations which maybe or can be computationally expensive and for 3D data the figures may increase exponentially.

2.2.3 Relation to the Project:

The relation of this paper to the project we are working on are:

- 1-The diffusion and denoising process used in this paper could also be used for Inpainting tasks, to fill out the voided images of brain MRI, and synthesis healthy tissues in region affected by glioma
- 2- For making sure the generated MRI image or the inpainted part is anatomically accurate we can use the semantic labelling used in this paper, also the diffusion model can ensure the structural coherence throughout the multiple iterations occurred.

2.3 Denoising Diffusion Models for 3D Healthy Brain Tissue Inpainting

2.3.1 Summary:

This paper [4] explores the DDPMs for the inpainting and its uses to generate the 3D images for the area effected by the disease or brain having the tumors. The main motivation is the need to restore the image of heathy area for the better diagnosis. The authors of this paper analyse and evaluate and modify different models including 2D, pseudo-3D, and fully 3D methods, to solve and overcome the challenge of consistent

inpainting in 3D MRIs. the main contribution through this paper and its study is the extension of 2D and 3D Diffusion models for inpainting and balancing the performance of pseudo 3D model as best approach.as this model outperform the traditional approach that was through the FSL Lesion-filing tool.

2.3.2 Critical analysis:

1-The paper directly works on the main task of our project we are working on glioma affected regions. DDPMs usage for this task is highly relevant, as DDPMs are used for iteratively denoise and generate tissue structures, which is important for filling in missing or damaged brain tissue. The BraTS dataset is also used in the paper.

2-The paper includes pseudo-3D method which is dealing with this issue by utilizing 2D convolutions and 1D convolutions in the z-axis, which ensures consistent generation of 3D volumes. This model outperform the 2D slice based and efficient in terms of SSIM, MSE and PSNR . Which makes it a good choice to use for brain inpainting task.

3-This paper also shows the clinical relevance of inpainting by testing the model on a MS lesion-filling task, where the pseudo-3D method outperforms the traditional FSL lesion filling tool, which indicates that this approach could be similarly effective for glioma affected regions.

4-The 3D diffusion models are computationally heavy, with sampling times reaching up to 20 m/vol, which makes them impractical for real-time applications. This could be a problem for our project where handling a large amount of data quickly is important..

2.3.3 Relation to our project:

Both of them are focused on almost same issue as they are working on reconstructing brain MRI after it got affected by the disease and reconstruct the healthy brain after it got affected by glioma. Also uses DDMPs for reaching the goal and which could also be helpful to us while choosing which model to go with.

2.4 Multitask Brain Tumor In-painting with Diffusion Models

2.4.1 Summary:

This paper [15] focus on the methodological approach of using Diffusion models[6] for the in-painting and multitasking brain tumor. And also focus on to improve the accuracy and efficiency of these in-painting methods specifically related to brain tumors. This paper proposed the multitask frameworks where the trained model is able to multitask, tasks like segmentation of tumor and tissue in-painting, which helps in improving the performance. The authors use the dataset of brain MRI s with annotated tumors by evaluating the performance of methods using same metrics we talked about earlier i.e. SSIM,PSNR. Few enhanced methods can help in improving diagnostic outcomes and support in clinical decision making by giving more accurate and clearer representation of brain tissues and tumors.

2.4.2 Critical analysis:

1-The use of diffusion models for multitask brain tumor inpainting is a novel approach. These diffusion models have capabilities in generating and in image tasking. Using them in inpainting in the multitask framework is innovative.

2-As these inpainting techniques will be improved this can directly benefits the clinical methods by enhancing the MRI interpretations and potentially aiding in a better monitoring.

3-While SSIM and PSNR are commonly used metrics, they may not be able to capture the clinical insignificant of the inpainting results. Use of new and additional metrics could help in improving and producing the best outcome.

4-This method was solely tested on a dataset from a Mayo clinic. Therefore, the generalizability of the method is need to be addressed. The method may behave differently due to the variability in MRI protocols and tumor characteristics. A critical analysis with the widely used inpainting methods can provide a clear image.

5-Further improvement can be brought by conducting experiments on the different dataset from different institutions and exploring methods that can reduce computational burden of diffusion model and making them more practical for the real time clinics.

2.4.3 Relation to our project:

The Research paper is using the multitask framework integrating tumor segmentation and tissue inpainting which align directly with the project we are working on, by training

the model to multitask which effectively captures and reconstruct the missing tissues that aligns with our main goals too. This paper is using the diffusion models which are frequently used in challenge of generating realistic and coherent brain tissue, preserving both global and local anatomical features which is key requirement in our project

2.5 Blind MRI Brain Lesion In-painting Using Deep Learning

2.5.1 Summary:

This research paper [12] focusses on an inpainting technique for brain MRI images damaged by lesions using deep learning. By using a 3D U-Net architecture for blind inpainting, the technique replaces the requirement for manual lesion delineation by automatically reconstructing parts of the lesion with brain tissue that seems healthy. The network is taught in two stages: first, artificial training data is created using a non-blind inpainting network, and then the inpainted images are utilised to train a blind inpainting network. The model improves robustness against the existence of lesions, which improves the performance of existing pipelines, such as brain tissue segmentation.

2.5.2 Critical Analysis:

- 1-The time-consuming and error-prone nature of conventional in-painting techniques is decreased by using blind in-painting, which eliminates the need for manual lesion area identification.
- 2-The use of a complete 3D U-Net, rather than 2D slices or 3D patches, allows for better context awareness and more accurate in-painting of entire brain volumes.
- 3-Given the solutions' possible clinical use, the computational efficiency of the model , which can handle data in as little as 0.5 seconds is remarkable.
- 4-As the method is primarily intended to treat white matter anomalies, it is not generalized to other brain tumors, such as gliomas or vascular lesions.

2.5.3 Relation to the project:

By concentrating on glioma-aware in-painting, our approach expands on previous research and tackles the difficulty of rebuilding brain tissue damaged by tumours instead of only white matter abnormalities. MRI and a larger range of assessment metrics, such as the Dice coefficient, are combined.

2.6 GAN-based metal artifacts region inpainting in brain MRI imaging with reflective registration

2.6.1 Summary:

This research paper [18] describes the metal Artifacts Region in-painting Network(MARINet), a deep learning model / for in-painting regions in brain MRI with metal artifacts . MARINet uses U-Net architecture with dual-path encoder and takes advantage of brain symmetry via reflective version of it [14] .The model in-paints metal artifacts by referring to both the original image and a flipped registered version of it.The research compares MARINet against existing in-painting approaches , including partial Convolution(PCony) and Gated Convolution(GCony), and finds that it outperforms them. Evaluation metrics such as mean absolute error(MAE) and peak signal-to-noise ratio(PSNR) demonstrate MARINet's ability to improve image quality and restore texture of tissue.The model is applied to clinical MRI scans with metal implants[17].The dataset used in this research paper includes:- Patient Data MRI images from 100 patients without dentures, scanned between 2017 and 2021. This group included 61 males and 39 females, with an average age of 57 years and Image Data:T1-weighted MRI images containing or located near the teeth, resulting in 9000 slices after data augmentation.

2.6.2 Critical Analysis:

1-MARINet uses reflective registration to utilise the quasi-symmetry of brain pictures, dramatically improving in-painting quality.

2-MARINet outperforms other approaches like as PConv and GConv in terms of numerous metrics (MAE, PSNR, and SSIM).

3-It solves a critical problem in MRI imaging, particularly in the presence of metal artefacts, by producing more realistic reconstructions of damaged/tumor-affected regions.

4-The dual-path encoder in U-Net, along with partial convolution, enables MARINet to focus on valid pixel areas, enhancing in-painting accuracy.

5-Because the model relies on brain symmetry, it may perform badly in circumstances of asymmetry, such as when artefacts exist on both sides of the brain.

6-The method focusses on 2D in-painting rather than fully utilising the 3D structure of MRI data.

2.6.3 Relation to the project:

MARINet's inpainting approach based on brain symmetry is directly relevant to our effort because both aim to restore MRI image regions impacted by artefacts. Our initiative focusses on tumor-affected regions, specifically gliomas, whereas MARINet focusses on metal artefacts. The reflecting registration technique and the utilisation of symmetry may inspire new approaches to treating glioma-affected areas, particularly in cases where brain symmetry remains largely intact.

2.7 Inpainting non-anatomical objects in brain imaging using enhanced deep convolutional autoencoder network

2.7.1 Summary:

This research paper [9] discusses an enhanced deep convolutional auto-encoder network for in-painting non-anatomical objects in brain MRI and CT scan data. The suggested method employs deep learning techniques, notably convolutional neural networks (CNNs) and generative adversarial networks (GANs), to restore missing or distorted portions in medical images. To improve pixel precision and ensure global consistency, the model employs an enhanced encoder-decoder architecture with skip connections, as well as two adversarial losses (global local). The method is faster and more precise than previous methods, resulting in high-quality findings for PSNR (52.44), SSIM (0.95), and MSE (40.73). It considerably improves diagnostic image quality, particularly in areas affected by non-anatomical aberrations such artifacts [16] or object removal. The dataset utilised in this study comes from the Visible Human Project (VHP). The pictures were created with Siemens Biograph mCT and downsampled to 1x1x1 mm from the original 0.85x0.85x5 mm. The dataset contains 800 photos, which were separated into three groups: training, testing, and validation, each of 640, 80, and 80 images [2].

2.7.2 Critical Analysis

1-The combination of adversarial losses make sure the realistic in-painting and better consistency in the reconstructed areas.

2-Preserve high-resolution feature information, improving the fidelity of the in-painted images.

3-The paper demonstrates excellent quantitative results in metrics like PSNR, SSIM, and MSE. The method accounts for both global and local losses, which enhances the realism of the final output.

4-The focus is on non-anatomical object removal rather than pathological tissue reconstruction. The method does not specifically address the unique challenges of in-painting regions affected by gliomas, such as irregular tumor boundaries or heterogeneous textures.

5-As the method is said to be faster, the use of two adversarial networks and high-resolution in-painting may still require significant computational resources.

2.7.3 Relation to the project:

This research work emphasises on non-anatomical in-painting [11], which is rather different from our project's goal of in-painting glioma-affected brain tissue. Certain essential components, such as the use of CNNs, skip connections, and adversarial losses, are compatible with our project's technical approach. The proposed model may inspire the creation of our own in-painting framework, especially in terms of preserving global consistency and high-resolution details in 3D brain MRI data.

2.8 Brain MR Atlas Construction Using Symmetric Deep Neural Inpainting

2.8.1 Summary:

This paper [19] describes a deep learning-based method for in-painting brain MRI data, focusing on tumor-affected areas. The method seeks to replace tumor regions with synthetic normal tissue to ease brain atlas building, which is a bit difficult when tumors are present. The proposed method constructs a statistical brain atlas using only pathological MRI data [5], filling irregularly shaped tumor patches with realistic tissue. Handling randomly dispersed tumor locations, leveraging a symmetry constraint between brain hemispheres, and making use of partial datasets are all examples of key

improvements. The method was tested on the Multimodal Brain Tumor Segmentation (BraTS) dataset and demonstrated higher reconstruction quality, cross-correlation, and mutual information metrics than earlier methods. The 2018 Multimodal Brain Tumor Segmentation (BraTS) dataset is used in this paper. This database consists of approximately 210 clinical 3T multimodal MRI scans of glioblastoma and lower-grade glioma with pathologically confirmed diagnoses. The tumor locations were manually identified by expert neuro-radiologists and used as ground truth. The dataset includes T1-weighted, T2-weighted, and T1-weighted contrast-enhanced (T1ce) MRI scans [1].

2.8.2 Critical Analysis:

1. The technique successfully handles incomplete datasets with irregular tumour regions, overcoming a common limitation in brain atlas creation.
2. The symmetry restriction between brain hemispheres improves in-painting accuracy, yielding more realistically reproduced regions.
3. Several assessment metrics (SSIM, PSNR, and L1 error) show that the method outperforms earlier in-painting techniques such as patch-match and partial convolution.
4. In this paper, the method relies mostly on 2D slices, restricting its potential to capture 3D information, which could increase in-painting accuracy in MR data.
5. The symmetry notion, while useful, may not always hold true due to natural asymmetry in brain structures, and significant misalignments might impair performance.
6. The method is strongly reliant on the manual modification of loss function weights, which may reduce in-painting performance and create bias.

2.8.3 Relation to the project:

This study closely resembles our project in that both aim to rebuild brain tissue in tumor affected areas. While this research focusses on statistical brain atlas building and uses 2D slices, our project will broaden its scope by focussing on 3D inpainting, making it more useful to clinical analysis tasks such as tumour monitoring and tissue segmentation. The symmetry restriction they applied can serve as inspiration for our method, but we want to take it a step further by combining multi-modal MRI data and advanced 3D inpainting techniques.

Comparative Table

Reference	Dataset	Data Preprocess	Accuracy	Evaluation Method	Pipeline	Weakness
Nature (2020)	680 T1-weighted 3D MR images from KBASE (527 for training, 153 for testing)	Patch-based learning PET/MRI preprocessing of 3D MRI	nil	PSNR, SSIM, HFEN	U-Net-like structure with DenseNet sub-blocks	Boundary artifacts, high computational cost
HAL (2020)	Open access IXI dataset MS dataset composed of 3T 3D-T1w MPAGE images	Non-blind inpainting followed by blind inpainting	PSNR: 48.13, Correlation: 0.9997	PSNR, DICE, correlation	3D U-Net architecture	Focus on white matter lesions, reliance on synthetic data, computational complexity
PMC (2022)	BraTS 2018 (210 glioma cases)	Symmetry constraint handling incomplete datasets	SSIM: 0.9823, PSNR: 35.85, Mean L1 error: 211.7, Inception score: 11.74	SSIM, PSNR, Mean L1 error, inception score	Coarse-to-fine architecture with patch-swap	Use of 2D slices, symmetry assumption may not always hold
Wiley (2024)	9000 slices from 100 patients with metal artifacts	Images were acquired on a Philips Achieva scanner using a T1-weighted high-resolution isotropic volume examination sequence. Reflective registration was performed on the corrupted image for the test and real clinical artifact data.	MAE: $18.34 \pm 2.75 \times 10^{-2}$, PSNR: 17.62 ± 0.96 , SSIM: <u>98.63 ± 0.46</u>	800 images from the Visible Human Project	Enhanced deep convolutional autoencoder, data downsampling	the model may perform poorly in cases where symmetry is not applicable. This approach required reflective registration, which may reduce effectiveness when artifacts are large or affect both sides of the brain.
DDPMs	BraTS 2023 containing T1weighted brain MR scans with tumors	Publicly available on BraTS website	SSIM, MSE, PSNR	0.8527 ± 0.1196 , 0.0103 ± 0.0107 , 20.9258, 3.3835	Expensive, results depend on quality and diversity of dataset. Lacks clinical	Not good results with 3D models

2. Preliminary Literature Review

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Table 2.1: Comparison of Various Methods and Datasets

Approximate methodology/Proposal Approach

3.1 Verifying the BraTS Challenge Dataset for Brain MRI

Inpainting Workflow

3.1.1 Verification of data integration:

- Download the BraTS dataset.
- Confirm the presence of ground truth segmentation masks for tumor regions.

3.1.2 Visual Inspection

• We chose a MRI volume sample at random. Then we superimposed the matching segmentations from each data set and visualized them. By this method we were able to recognize all the corrupted files and checked whether the pictures align with the data set correctly.

• Corrective measures, like resampling or removing faulty data from the dataset, are implemented when the problems of alignment and corrupted files were encountered.

3.1.3 Statistical Consistency

- We calculated **statistical measures** such as mean, variance, and intensity histograms for MRI scans. This helped in finding any sort of anomalies from the data set and outliers
- When the anomalies were found in the data set like very high or low values. Then the normalization techniques were carried out.

3.2 Workflow for Rescaling, Normalization, and Ensuring Consistent Voxel Size

Once we validated the dataset. Our next step was to preprocess whether the images are suitable for the inpainting tasks.

3.2.1 Rescaling to Consistent Voxel Size

- MRI scans often varied in voxel resolutions depending upon which sort of scanner is being used. In this step, we rescaled all the MRI volumes to a **consistent voxel size of 1x1x1 mm³**.
- **3D interpolation** (such as linear or cubic interpolation) is used to achieve this rescaling. This helps ensuring that anatomical structures are preserved and that voxel size discrepancies do not impact the model's ability to reconstruct brain tissue.
- Our final goal was to see the images consistency which will help in the model training.

3.2.2 Normalization of Intensity Values

- MRI scans may have different ranges which may confuse the deep learning model. To implement standardized ranges, we apply **Z-score normalization** for each modality. This step involves subtracting the mean voxel intensity and dividing by the standard deviation.
- In some cases, **percentile normalization** is used when the intensity range is normalized between the **1st and 99th percentiles** to [0, 1] or [-1, 1]. This helps in ensuring that learning process is not disturbed.

3.2.3 Ensuring Voxel Size Consistency

- After the data is rescaled, it was verified that all MRI volumes have **consistent voxel sizes** by checking the voxel size metadata stored in the image headers (DICOM/NIFTI).
- In case of any mismatches found, we resampled the affected images using **3D interpolation techniques** such as this was provided by image processing libraries like **SimpleITK** or **ANTs**.
- Second time the visualization technique was conducted to ensure that **anatomical fidelity** is maintained, and during resampling no distortions were introduced

We see a raw MRI scan with varying intensities and voxel sizes on the left side. The middle shows the transformation through **3D interpolation** for rescaling and **Z-score normalization**. While on the right side we have a uniformed intensity image with uniform voxel sizes. The above visual workflow emphasizes the necessity of preprocessing steps for maintaining consistency.

3.3 Quality Control and Final Validation

When the rescaling, normalization and voxel size corrections were done. Following steps were taken:

- **Random visual checks** to ensure that the inpainted regions are anatomically plausible.
- **Statistical checks** were used to compare the pre-processed dataset with the original dataset to check intensity distributions have not been distorted during preprocessing.
- We used the **Dice score** and **SSIM** to compare images pre-processing and post-processing, while ensuring that data intensity remains high.

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